UNIVERSITY OF DERBY
College of Science and Engineering

Practice Paper A

OPERATIONAL RESEARCH
5CM522

DATE: REVISION MATERIAL
TIME ALLOWED: 4 HOURS

Instructions to Candidates

You are required to answer a total of 4 questions.

However, you may attempt all 6 and your best 4 will be considered.

This is a practice paper designed to support preparation for the timed in-class test.

You may use Excel or Python, but your workings, files and interpretations must be clear and well labelled.

You may access code, notes and module files while practising.

You must not use AI tools if you want the practice to reflect the conditions of the real in-class test.

1.

Elmbridge Furniture makes two products each week: desks and shelving units.

Each desk uses 5 carpentry hours and 2 finishing hours. Each shelving unit uses 3 carpentry hours and 4 finishing hours. At most 120 carpentry hours and 96 finishing hours are available each week. At least 8 shelving units must be produced to satisfy an online retailer.

The weekly contribution is £58 per desk and £46 per shelving unit.

(a) Formulate a linear programming model for this problem, clearly defining the decision variables, objective function and constraints. (6 Marks)

(b) Solve the model to determine the weekly production plan that maximises contribution. You may use a graphical method, algebra or Python, but your work must be clear. (6 Marks)

(c) Identify the binding constraints and determine the slack in each constraint at the optimum. (4 Marks)

(d) The contribution on shelving units rises to £52. Determine whether the optimal basis changes and justify your answer clearly. (5 Marks)

(e) A new agreement requires shelving units to make up at least 30% of total weekly output. Reformulate the model and determine whether the original optimum remains feasible. If not, find the revised optimal plan. (4 Marks)

Total 25 Marks

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2.

GreenGrid Energy is choosing a set of improvement projects for a regional depot network. Each project is either undertaken in full or not undertaken at all.

Project	Cost (£m)	Annual Benefit Score
A	6	14
B	8	17
C	5	11
D	9	20
E	4	9
F	7	15

The total budget is £21 million.

The following additional conditions apply:

- if project D is selected, project B must also be selected;
- projects A and F cannot both be selected;
- at least one of projects C and E must be selected.

(a) Formulate a binary integer programming model to maximise total annual benefit subject to the budget and project-selection rules. Clearly define all decision variables. (8 Marks)

(b) Solve the model in Python or by systematic analysis, and determine which projects should be selected and the maximum total annual benefit score. (7 Marks)

(c) Explain why this problem requires binary decision variables rather than the continuous variables used in ordinary linear programming. (3 Marks)

(d) Modify the model so that exactly four projects must be selected, then resolve or explain how the optimal solution would change. (4 Marks)

(e) Suggest one realistic extension that would make this capital-selection model more useful in practice. No full formulation is required. (3 Marks)

Total 25 Marks

3.

The network below represents proposed communication links between seven depots. The number on each edge is the installation cost in thousands of pounds.

Link	Cost	Link	Cost
A–B	6	C–F	6
A–C	4	D–E	5
A–D	9	D–F	7
B–C	5	D–G	8
B–E	7	E–F	3
C–D	6	E–G	4
C–E	8	F–G	5

(a) Explain what is meant by a minimum spanning tree and why it is useful in network design. (4 Marks)

(b) Use either Prim's algorithm or Kruskal's algorithm to determine a minimum spanning tree for this network. Show the main steps of the algorithm. (10 Marks)

(c) State the selected links and the total installation cost of the spanning tree. (4 Marks)

(d) The firm now requires one additional backup link so that there is exactly one cycle in the final design. Identify the cheapest such addition and state the revised total cost. (4 Marks)

(e) Comment briefly on one limitation of using only an MST for a real logistics or telecoms network. (3 Marks)

Total 25 Marks

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4.

Westmoor Cars is deciding how many display bays to provide for newly delivered vehicles. Each week, the number of cars delivered and the number of cars requested by customers are uncertain.

Cars Delivered	6	7	8	9	10
Probability	0.10	0.20	0.35	0.25	0.10

Weekly Demand	5	6	7	8	9
Probability	0.15	0.25	0.30	0.20	0.10

Each permanent display bay costs £18 per week. Any cars that cannot be placed in a display bay must be held in overflow storage at a cost of £30 per car per week.

For consistency, use the following random numbers for a 12-week simulation:

Week	1	2	3	4	5	6	7	8	9	10	11	12
Delivery RN	0.119	0.503	0.512	0.860	0.103	0.223	0.601	0.557	0.783	0.548	0.731	0.768
Demand RN	0.751	0.587	0.240	0.614	0.111	0.817	0.450	0.815	0.685	0.679	0.210	0.252

(a) Describe how you would model this problem as a simulation study. Your answer should identify the stochastic inputs, state variables and performance measures. (5 Marks)

(b) Simulate 12 weeks of operation for a capacity of 7 display bays using the random numbers above. You may use Excel or Python to carry out the calculations, but your logic must be clear. (8 Marks)

(c) Estimate the average weekly cost for a capacity of 7 bays. (3 Marks)

(d) Repeat the analysis for capacities of 6, 8 and 9 bays using the same random numbers, then recommend a capacity. Present your results in a clearly labelled table. (6 Marks)

(e) State two reasons why a much longer simulation or repeated runs would improve confidence in the recommendation. (3 Marks)

Total 25 Marks

5.

The demand analyst at Calder Kitchens has collected the following quarterly sales data for one of its product ranges.

Period	Year	Quarter	Demand
1	2020	Q1	510
2	2020	Q2	620
3	2020	Q3	570
4	2020	Q4	690
5	2021	Q1	548
6	2021	Q2	662
7	2021	Q3	608
8	2021	Q4	742
9	2022	Q1	590
10	2022	Q2	708
11	2022	Q3	646
12	2022	Q4	792
13	2023	Q1	636
14	2023	Q2	760
15	2023	Q3	692
16	2023	Q4	852

(a) Using periods 1–12 only, calculate suitable centred 4-quarter moving averages and estimate the trend where possible. Plot the original data together with the estimated trend. (8 Marks)

(b) Using periods 1–12 only, calculate seasonal indices using both additive and multiplicative decomposition. (7 Marks)

(c) Use periods 13–16 as a validation set. Produce forecasts for those periods using both models and compare their accuracy using one suitable error measure. (6 Marks)

(d) Using the better model, forecast the four quarters of 2024 and justify your model choice briefly. (4 Marks)

Total 25 Marks

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6.

Northfield Relief ships medical supplies from three depots to four clinics through two regional hubs. Supplies may travel only from a depot to a hub and then from a hub to a clinic.

Depot	Supply	Cost to H_1	Cost to H_2
D_1	100	5	7
D_2	90	4	6
D_3	110	6	3

Hub	Cost to C_1	Cost to C_2	Cost to C_3	Cost to C_4	Total Demand
H_1	4	6	5	7	
H_2	6	4	3	5	
Demand	60	70	80	75	285

(a) Formulate a transhipment model for this problem. Clearly define the decision variables, objective function, supply constraints, demand constraints and hub flow-balance constraints. (8 Marks)

(b) Solve the model and determine the optimal shipment plan, the total transportation cost and any unused supply. Report only the non-zero shipments. (7 Marks)

(c) A disruption means that hub H_1 can process at most 130 units, and the route from H_2 to clinic C_3 is unavailable. Show how the model changes and determine the revised total cost. State only the revised non-zero shipments or the flows that change. (6 Marks)

(d) In the disrupted scenario, up to 20 extra units of processing capacity can be bought for hub H_1 at a one-off cost of £50. Determine whether this is worthwhile and justify your answer. (4 Marks)

Total 25 Marks

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